1

$X_i \overset{\text{iid}}{\sim} Gamma(\theta, k), k$ is known

I'm assuming this is for an $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$ test.

(a) Find asymptotic level $1 - \alpha$ confidence inerval for θ by inverting acceptance region of an asymptotic size α LRT

$L(\theta|x, k) = \prod \frac{k^\theta}{\Gamma(\theta)} x^{\theta-1} e^{-kx} = \frac{k^{n\theta}}{\Gamma(\theta)^n} x^{n\theta-n} e^{-k\sum x}$

Under null, $\hat{\theta}_R = \theta_0$

LRT: $\lambda(x) = \frac{L(\hat{\theta}_R|x,k)}{L(\hat{\theta|x,k})} =$

What is $\hat{\theta}$??? No closed form for the MLE for θ.

On this part we can develop an acceptance region and inverted test by just assuming we know the MLE of θ. However, to actually find an interval we would need to examine the LRT to see how it behaves

(b)

$\mathbb{E}[X] = \alpha\theta$. $M_1 = \bar{X}$

$\theta_{MoM} : \alpha\theta = \bar{X} \implies \theta_{MoM} = \frac{\bar{x}}{k}$

According to CLT, $\sqrt{n}(\bar{X} - k\theta) \sim N(0, \theta k^2)$.

$\implies$ by Cont. Map. Theorem: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{D} N(0, \theta)$

Variance is θ

(c)

We need $g'(\theta)^2 = \frac{1}{\theta} \implies g'(\theta) = \sqrt{\frac{1}{\theta}}$. Thus, $g(\theta) = 2\sqrt{\theta}$

(d)

$[g^{-1}(g(\hat{\theta}) - \frac{1}{\sqrt{n}} Z_{1-\alpha/2}), g^{-1}(g(\hat{\theta}) + \frac{1}{\sqrt{n}} Z_{1-\alpha/2})]$

$P(-Z_{\alpha/2} < \sqrt{n}(2\sqrt{\hat{\theta}} - 2\sqrt{\theta}) < Z_{\alpha/2}) \approx 1 - \alpha$

$P(\frac{-Z_{\alpha/2}}{2\sqrt{n}} \leq \sqrt{\hat{\theta}} - \sqrt{\theta} \leq \frac{Z_{\alpha/2}}{2\sqrt{n}})$

$(\sqrt{\hat{\theta}} - \frac{Z_{\alpha/2}}{2\sqrt{n}})^2 \leq \theta \leq (\sqrt{\hat{\theta}} + \frac{Z_{\alpha/2}}{2\sqrt{n}})^2$

2

$f(x|k, \theta) = \frac{\theta k^\theta}{x^{\theta+1}} = \theta(\frac{k}{x})^\theta \frac{1}{x}$

$L(k, \theta|x) = \prod \frac{\theta k^\theta}{x^{\theta+1}} = \frac{\theta^n k^{n\theta}}{\prod x^{\theta+1}}$

(a) θ is known, $H_0 : k = 1, H_1 : k = 2$. Find most powerful size α test.

We can ignore the case when $x < 1$ because under both hypothesis this is not possible.

LRT: $\frac{L(k=1|x,\theta)}{L(k=2|x,\theta)} = \frac{\frac{\theta^n 1^{n\theta}}{\prod x^{\theta+1}} I(x>2)}{\frac{\theta^n 2^{n\theta}}{\prod x^{\theta+1}} I(x>1)} = (\frac{1}{2})^{n\theta} I(X_{(1)} \geq 2)$

$$\lambda(x) = \begin{cases} 0, & X_{(1)} \leq 2 \\ (\frac{1}{2})^{n\theta}, & X_{(1)} > 2 \end{cases}$$

$$\phi(x) = \begin{cases} 1, & \lambda(x) > k \\ \gamma, & \lambda(x) = k \\ 0, & \lambda(x) < k \end{cases}$$

$\alpha = P(\lambda(x) > k) + \gamma P(\lambda(x) = k)$

$P(\lambda(x) > k) = P(X_{(1)} > 2)$. We go from the test function to the LRT. LRT can only be greater than some k when the LRT provides a value, which only happens when the min is greater than 2.

If we pick k = 0, the second term is also active. If $k = 2^{n\theta}$ then only the second term is active.

$P(X_{(1)} > 2) = P(x_1 > 2)^n = [\int_2^\infty \theta x^{-\theta-1} dx]^n = [-x^{-\theta}|_2^\infty]^n = 2^{-n\theta}$

We cannot get a set α test using this specific test, so we will need the γ term meaning we must let k be one of the boundary values.

(b) k is known. $H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$

Need UMP size α test.

Check for MLR: $T = \sum ln(\frac{X_i}{k})$

$\theta_2 > \theta_1 : \frac{L(\theta_2)}{L(\theta_1)} = (\frac{\theta_2}{\theta_1})^n e^{-T(\theta_2-\theta_1)}$. This is monotonically decreasing in θ_2.

Karlin-Rubin: $\phi(x) = \begin{cases} 1, & T < c \\ 0, & T > c \end{cases}$. Choose c such that $P_{\theta_0}(T < c) = \alpha$.

3

$f(x|\lambda) = \frac{2x}{\lambda^2} e^{-\frac{x^2}{\lambda^2}}; x, \lambda > 0$

$L(\lambda|x) = \prod 2x \frac{1}{\lambda^{2n}} e^{-\frac{1}{\lambda^2}\sum x^2}$

(a) T = $\sum X_i^2$. Show it's complete and sufficient.

$f(x|\lambda) = [\prod 2x][\frac{1}{\lambda^{2n}} e^{-\frac{1}{\lambda^2}T}] = h(x)g(T, \lambda)$ thus by factorization theorem this is sufficient. Because k is known, this is actually part of the exponential family (also can be shown by writing it as $h(x)c(\lambda)e^{w(\lambda)t(x)}$). Because this is part of the exponential family, and the support contains an open set, this is a complete statistic as well.

Still not too sure how to do pivots. This is a scale family, so the pivot would be something like λx, but I'm not sure if that's correct or how to proceed.

$T(X) = \sum x^2 : Y = X^2 \implies Y \sim Exp(\lambda^2) \implies \sum x^2 \sim Gamma(n1/\lambda^2)$

Scale family, need to adjust by a multiple of λ^2.

$Q = \frac{\sum X^2}{\lambda^2} \sim Gamma(n, 1)$

$P(a \leq Q \leq b) = 1 - \alpha$

$P(\sqrt{\frac{\sum X^2}{b}} \leq \lambda \leq \sqrt{\frac{\sum X^2}{a}})$

Those two equations give us our system to solve and allow us to find the minimum width (Lagrangians)

(b) $H_0 : \lambda = \lambda_0, H_1 : \lambda \neq \lambda_0$

$l(\lambda|x) = \sum ln(2x) - 2nln(\lambda) - \frac{\sum x^2}{\lambda^2}$

$\frac{\partial l}{\partial \lambda} = -\frac{2n}{\lambda} + \frac{2\sum x^2}{\lambda^3} = 0 \implies 2n\lambda^2 = 2\sum x^2 \implies \hat{\lambda} = \sqrt{\frac{\sum x^2}{n}} = \sqrt{\frac{T}{n}}$

LRT: $\frac{L(\lambda_0|x)}{L(\hat{\lambda|}x)} = \frac{\prod 2x \frac{1}{\lambda_0^{2n}} e^{-\frac{1}{\lambda_0^2}\sum x^2}}{\prod 2x \frac{1}{\sqrt{T/n}^{2n}} e^{-\frac{1}{\sqrt{T/n}^2}\sum x^2}} = (\frac{\sqrt{T/n}}{\lambda_0})^{2n} e^{\sum x^2 (\frac{n}{T} - \frac{1}{\lambda_0^2})}$

This is monotonically increasing in T, thus the rejection region of $a \leq \lambda(x) \leq b$ is equivalent to saying $a^* \leq T(x) \leq b^*$